

Career Paths & Course Catalogue

Strategic Analysis — 2025-2026

Prepared: June 07, 2026

Executive Summary

Metric	Value
Total career paths	10
Total unique courses	36
Shared courses (2+ paths)	5
Patch 1 paths (launch-ready)	5 — AI Engineer, DevOps, Cloud, Data Engineer, Full Stack Java
Patch 2 paths	5 — Frontend, MERN, Data Scientist, Security, Mobile
Avg courses per path	4.5 (down from 5.2 before optimisation)
Estimated Patch 1 gen cost	~22 courses × \$0.03 avg = ~\$0.66 (Haiku model)

Industry Demand Rankings — All 10 Paths

Rankings based on job-posting volume (LinkedIn/Indeed 2025 Q1), YoY salary growth, learner enrolment trends and strategic ROI for the platform.

Rank	Career Path	Difficulty	Duration	Demand	Salary Range (USD)	Patch
1	AI Engineer	Advanced	8 months	★★★★★	\$145–195K	1 ■
2	DevOps Engineer	Advanced	5 months	★★★★★	\$130–175K	1 ■
3	Cloud Engineer	Advanced	6 months	★★★★½	\$135–180K	1 ■
4	Data Engineer	Advanced	6 months	★★★★½	\$130–170K	1 ■
5	Full Stack Java Developer	Advanced	7 months	★★★★	\$120–160K	1 ■
6	Frontend Engineer	Intermediate	4 months	★★★★	\$100–140K	2 ■
7	MERN Stack	Intermediate	5 months	★★★★	\$100–140K	2 ■
8	Data Scientist	Intermediate	6 months	★★★½	\$115–155K	2 ■
9	Web & Cloud Security Engineer	Advanced	6 months	★★★★	\$130–175K	2 ■
10	Mobile Developer	Intermediate	5 months	★★★	\$105–145K	2 ■

Key insight: Patch 1 focuses on infrastructure + AI roles where job-market demand is highest and salary premiums are strongest. Patch 2 covers high-learner-volume tracks where content overlap reduces net production cost

significantly.

Patch 1 — Launch-Priority Career Paths

These 5 paths should be generated and published first. They target the highest-paying, fastest-growing roles and require 22 unique course topics (some shared).

1. AI Engineer ● Advanced · 8 months · \$145–195K

Fastest-growing tech role in 2025–26. LLM-native tooling is mainstream; demand for production AI outpaces supply 3:1.

#	Course	Category	Difficulty
1	Python for AI/ML	AI & Machine Learning	Beginner
2	Large Language Models	AI & Machine Learning	Intermediate
3	Retrieval-Augmented Generation	AI & Machine Learning	Intermediate
4	AI Agents & Agentic Workflows	AI & Machine Learning	Advanced
5	MLOps & Model Deployment	AI & Machine Learning	Advanced

2. DevOps Engineer ● Advanced · 5 months · \$130–175K

Evergreen demand. Platform engineering is the new DevOps title; Kubernetes + Terraform are must-haves for every cloud hire.

#	Course	Category	Difficulty
1	Linux & Shell Scripting	DevOps	Beginner
2	Docker & Containers	DevOps	Beginner
3	Kubernetes	DevOps	Intermediate
4	Terraform & Infrastructure as Code	DevOps	Intermediate
5	CI/CD with GitHub Actions	DevOps	Beginner

3. Cloud Engineer ● Advanced · 6 months · \$135–180K

AWS holds 33 % cloud market share; SAA-C03 remains the #1 entry-level cloud cert by job-listing volume.

#	Course	Category	Difficulty
1	AWS Core Services	Cloud	Beginner
2	AWS Solutions Architect	Cloud	Intermediate
3	Terraform & Infrastructure as Code	DevOps	Intermediate
4	Cloud Security	Cloud	Intermediate
5	Serverless Architecture	Cloud	Intermediate

4. Data Engineer ● Advanced · 6 months · \$130–170K

Data pipelines underpin every AI initiative. dbt + Airflow + Spark is now the standard modern data stack hired by 70 % of data teams.

#	Course	Category	Difficulty
1	SQL Mastery	Data Engineering	Beginner
2	Apache Kafka & Messaging	Data Engineering	Intermediate
3	Apache Spark	Data Engineering	Intermediate

4	Apache Airflow & Orchestration	Data Engineering	Intermediate
5	dbt & Analytics Engineering	Data Engineering	Intermediate

5. Full Stack Java Developer ● Advanced · 7 months · \$120–160K

Java remains the dominant enterprise language. Spring Boot + React is the standard stack at banks, fintechs and large ISVs.

#	Course	Category	Difficulty
1	Java Spring Boot	Backend	Intermediate
2	PostgreSQL Mastery	Backend	Intermediate
3	React.js	Frontend	Beginner
4	Docker & Containers	DevOps	Beginner

Patch 2 — Second-Wave Career Paths

These 5 paths leverage high content-overlap with Patch 1 tracks, cutting net production cost. Frontend, MERN and Mobile share TypeScript and React with Full Stack Java and MERN.

6. Frontend Engineer ● Intermediate · 4 months · \$100–140K

React + Next.js + TypeScript is the dominant frontend trio. Strong overlap with MERN and Full Stack Java paths reduces content cost.

#	Course	Category	Difficulty
1	TypeScript	Frontend	Beginner
2	React.js	Frontend	Beginner
3	Next.js	Frontend	Intermediate
4	Tailwind CSS & Modern CSS	Frontend	Beginner

7. MERN Stack ● Intermediate · 5 months · \$100–140K

Most popular "first full-stack" learning path globally. 3 of 4 courses shared with Frontend/Full Stack Java reduces net cost.

#	Course	Category	Difficulty
1	React.js	Frontend	Beginner
2	Node.js & Express	Backend	Beginner
3	TypeScript	Frontend	Beginner
4	MongoDB	Backend	Beginner

8. Data Scientist ● Intermediate · 6 months · \$115–155K

Strong learner demand but role is evolving toward AI Engineer. Ranked 8th to allow Patch 1 AI Engineer track to launch first.

#	Course	Category	Difficulty
1	Python for AI/ML	AI & Machine Learning	Beginner
2	SQL Mastery	Data Engineering	Beginner
3	Machine Learning Fundamentals	AI & Machine Learning	Beginner
4	Data Science with Python	AI & Machine Learning	Beginner

9. Web & Cloud Security Engineer ● Advanced · 6 months · \$130–175K

Defensive/application security scope only (renamed from Cybersecurity). High industry demand but niche learner base; Patch 2 placement is optimal.

#	Course	Category	Difficulty
1	Linux & Shell Scripting	DevOps	Beginner
2	Web App Security OWASP	Security	Intermediate
3	API Security	Security	Intermediate
4	Cloud Security Fundamentals	Security	Beginner
5	DevSecOps	DevOps	Intermediate

10. Mobile Developer ● Intermediate · 5 months · \$105–145K

Healthy but slower-growth segment. React Native + Flutter dual coverage maximises reach. TypeScript shared with Frontend/MERN paths.

#	Course	Category	Difficulty
1	TypeScript	Frontend	Beginner
2	React Native	Frontend	Intermediate
3	Flutter	Frontend	Intermediate
4	API Design & Best Practices	System Design	Intermediate

Course Sharing Map

Courses appearing in multiple paths are generated once and reused, reducing cost and ensuring consistency. Shared courses are shown in **bold** in the path detail above.

Course	Used in Paths	Count
TypeScript	Frontend, MERN, Mobile	3
React.js	Frontend, MERN, Full Stack Java	3
Docker & Containers	DevOps, Full Stack Java	2
Terraform & Infra as Code	DevOps, Cloud	2
Python for AI/ML	AI Engineer, Data Scientist	2
Linux & Shell Scripting	DevOps, Security	2
SQL Mastery	Data Engineer, Data Scientist	2
Cloud Security	Cloud, Security	2
Next.js	Frontend	1
Tailwind CSS	Frontend	1
Node.js & Express	MERN	1
MongoDB	MERN	1
Spring Boot	Full Stack Java	1
PostgreSQL Mastery	Full Stack Java	1
Kubernetes	DevOps	1
AWS Core / Solutions Arch	Cloud	1
Serverless Architecture	Cloud	1
LLMs	AI Engineer	1
RAG	AI Engineer	1
AI Agents	AI Engineer	1
MLOps	AI Engineer	1
Kafka	Data Engineer	1
Apache Spark	Data Engineer	1
Airflow	Data Engineer	1
dbt	Data Engineer	1
ML Fundamentals	Data Scientist	1
Data Science with Python	Data Scientist	1
Web App Security OWASP	Security	1
API Security	Security	1
DevSecOps	Security	1
React Native	Mobile	1
Flutter	Mobile	1

API Design	Mobile	1
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Full Course Catalogue — 36 Courses

Complete list of all courses in the platform catalogue, organised by career path coverage only.

#	Course Name	Category	Difficulty
1	Python for AI/ML	AI & ML	Beginner
2	Large Language Models	AI & ML	Intermediate
3	AI Agents & Agentic Workflows	AI & ML	Advanced
4	Retrieval-Augmented Generation	AI & ML	Intermediate
5	MLOps & Model Deployment	AI & ML	Advanced
6	Data Science with Python	AI & ML	Beginner
7	Machine Learning Fundamentals	AI & ML	Beginner
8	AWS Core Services	Cloud	Beginner
9	AWS Solutions Architect	Cloud	Intermediate
10	Cloud Security	Cloud	Intermediate
11	Serverless Architecture	Cloud	Intermediate
12	Docker & Containers	DevOps	Beginner
13	Kubernetes	DevOps	Intermediate
14	Terraform & Infrastructure as Code	DevOps	Intermediate
15	CI/CD with GitHub Actions	DevOps	Beginner
16	Linux & Shell Scripting	DevOps	Beginner
17	DevSecOps	DevOps	Intermediate
18	Node.js & Express	Backend	Beginner
19	Java Spring Boot	Backend	Intermediate
20	Apache Kafka & Messaging	Backend	Intermediate
21	PostgreSQL Mastery	Backend	Intermediate
22	SQL Mastery	Data Engineering	Beginner
23	Apache Spark	Data Engineering	Intermediate
24	dbt & Analytics Engineering	Data Engineering	Intermediate
25	Apache Airflow & Orchestration	Data Engineering	Intermediate
26	React.js	Frontend	Beginner
27	Next.js	Frontend	Intermediate
28	TypeScript	Frontend	Beginner
29	Tailwind CSS & Modern CSS	Frontend	Beginner
30	React Native	Frontend	Intermediate
31	Flutter	Frontend	Intermediate
32	API Design & Best Practices	System Design	Intermediate
33	Web App Security OWASP	Security	Intermediate
34	API Security	Security	Intermediate
35	Cloud Security Fundamentals	Security	Beginner
36	MongoDB	Backend	Beginner

Difficulty colour legend: **Beginner** **Intermediate** **Advanced**

Generation Cost Estimate

Using Claude Haiku at \$0.80/M input + \$4/M output tokens. Each lesson: ~2,000 input tokens + ~3,500 output tokens = ~\$0.0156/lesson. 35 lessons per course.

Patch	Unique Courses	Lessons	Est. Cost	Notes
Patch 1	22	770	\$12.00	AI, DevOps, Cloud, Data Eng, Java — no prior content
Patch 2	10*	350	\$5.46	*Excludes 13 courses shared with Patch 1
Total	32	1,120	\$17.46	Remaining 29 catalogue courses for future phases

Note: 8 AI/ML courses (indexes 0–11 minus Data Science with Python) have already been generated in a prior run at ~\$4.42. The remaining 22 Patch-1 courses are the priority generation target.